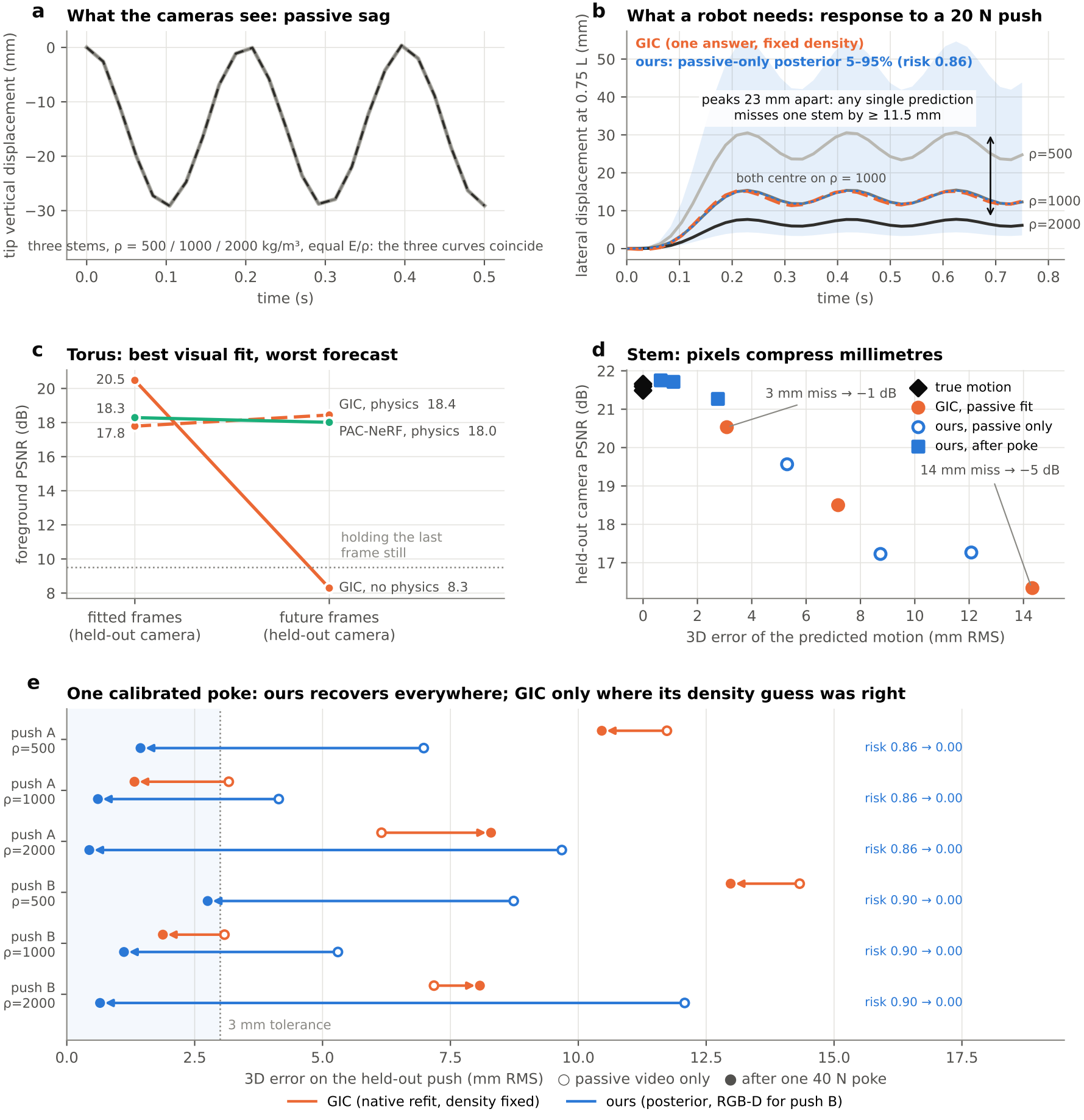
a
What the cameras see: passive sag
tip vertical displacement (mm)
0
−10
−20
−30
three stems, ρ = 500 / 1000 / 2000 kg/m³, equal E/ρ: the three curves coincide
0.0
0.1
0.2
0.3
0.4
0.5
time (s)
b
What a robot needs: response to a 20 N push
GIC (one answer, fixed density)
ours: passive-only posterior 5–95% (risk 0.86)
peaks 23 mm apart: any single prediction
misses one stem by ≥ 11.5 mm
both centre on ρ = 1000
ρ=500
ρ=1000
ρ=2000
lateral displacement at 0.75 L (mm)
50
40
30
20
10
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
time (s)
c
Torus: best visual fit, worst forecast
foreground PSNR (dB)
20.5
18.3
17.8
GIC, physics 18.4
PAC-NeRF, physics 18.0
holding the last
frame still
GIC, no physics 8.3
20
18
16
14
12
10
8
fitted frames
(held-out camera)
future frames
(held-out camera)
d
Stem: pixels compress millimetres
held-out camera PSNR (dB)
3 mm miss → −1 dB
14 mm miss → −5 dB
true motion
GIC, passive fit
ours, passive only
ours, after poke
22
21
20
19
18
17
0
2
4
6
8
10
12
14
3D error of the predicted motion (mm RMS)
e
One calibrated poke: ours recovers everywhere; GIC only where its density guess was right
push A
ρ=500
push A
ρ=1000
push A
ρ=2000
push B
ρ=500
push B
ρ=1000
push B
ρ=2000
risk 0.86 → 0.00
risk 0.86 → 0.00
risk 0.86 → 0.00
risk 0.90 → 0.00
risk 0.90 → 0.00
risk 0.90 → 0.00
3 mm tolerance
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
3D error on the held-out push (mm RMS)
passive video only
after one 40 N poke
GIC (native refit, density fixed)
ours (posterior, RGB-D for push B)